

Chinese checkers

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Time Limit: 1 s Memory Limit: 64 MB
Submission: 43 AC: 18 Score: 98.93

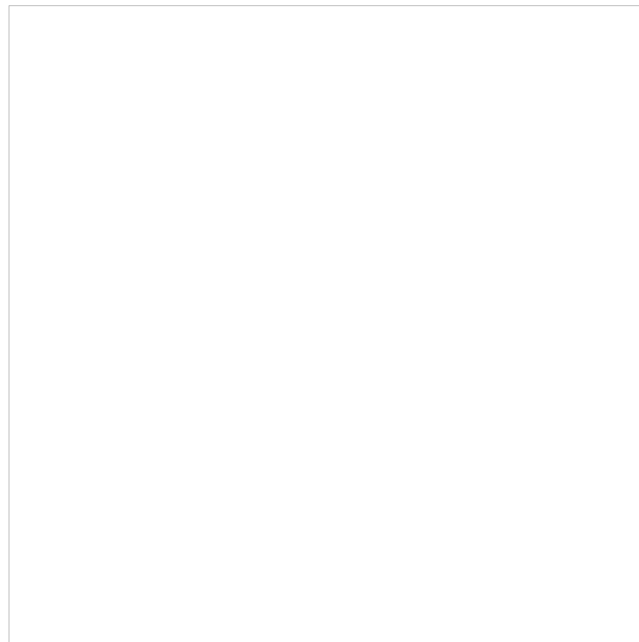
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Codes

AI Code Tips

Description

I think almost everyone play Chinese checkers when we were young.



Recently, hft777 is fond of playing Chinese checkers. But he is puzzled with how many steps at least to move all his pieces to his opposite position.

To make the problem simple, now we take a line into account. There are n positions on a line, numbered from 0 to $n-1$. At first, two same balls are on 0 and 1 position, and every time you can move a ball left or right, or you can move a ball on the position x to the position z on the other side of the ball on the position y , and $|x-y|=|y-z|$. And you must make sure the ball can't move out of the boundary, and the two balls can't be on one same position. Now, hft777 wants to know how many steps at least to move the two same balls to the $n-2$, $n-1$ positions.

Input

The first line of the input is one integer t , the number of testcases.

For each testcase, one integer n for a line, corresponding to the length of the line. $2 \leq n < 1000$

Output

For each testcase output the minimum steps.

Samples

input	Copy
<pre>2 3 4</pre>	
output	Copy
<pre>1 2</pre>	

Source

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